

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Social Network Analysis [NPT-4001]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given the following graph represented by its adjacency list, identify all connected components.

- A)
- Node 1: [2]
 - Node 2: [1, 3]
 - Node 3: [2]
 - Node 4: [5]
 - Node 5: [4]
 - Node 6: []
- (5)

- B) Explain the concept of Egocentric network? Illustrate a 1.5 egocentric network centered around an ego node 6 and 8.

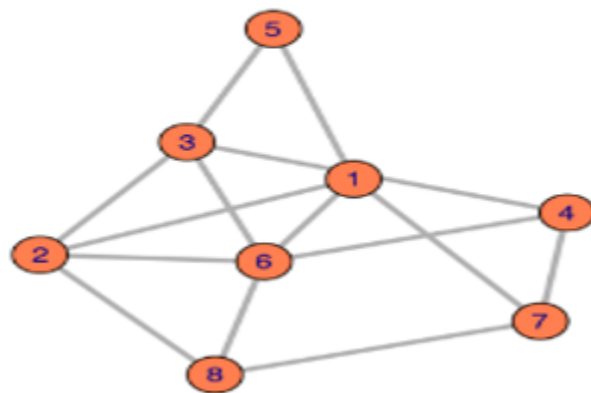


Figure 1: The Graph for reasoning

- 2) Explain the concept of clustering coefficient and average path length. How do these metrics reflect the properties of small-world networks?

(5)

(5)

A)

B)

(5)

Evaluate the density of the graph with respect to node A. Explain the density of the graph in terms of directed and undirected network.

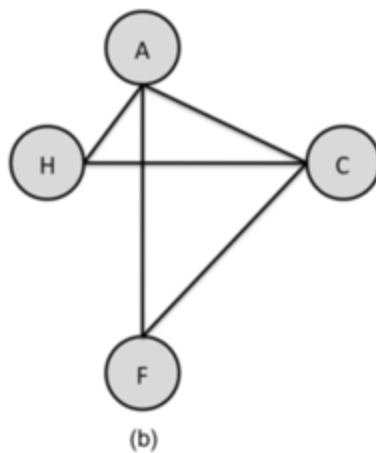


Figure 1: Graph for evaluation

- 3) Based on the connected pairs in the graph below, please complete the table mentioned below with scores using the specified link prediction formula.

A)

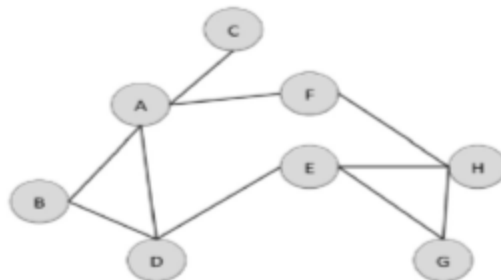
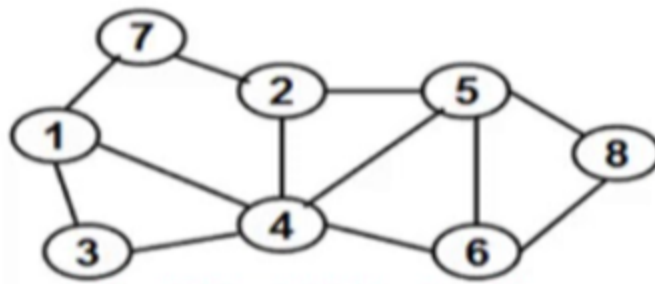


Figure 5: The Graph for reasoning

Pair	Common Neighbours	Jaccard Index	Preferential Attachment
D, G	-	-	-
B, E	-	-	-
E, F	-	-	-

B)

Illustrate the difference between Jaccard coefficient and Adamic /Adar method. Compute the most likely new edges using Jaccard and Adamic/Adar method.



4)

(5)

A)

Compute Laplacian Matrix and Normalized Laplacian matrix for this graph. Illustrate the properties of Laplacian Matrix.

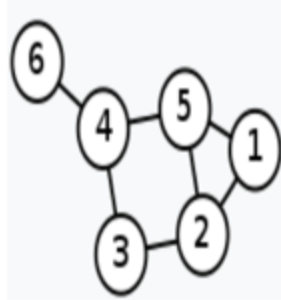


Figure 4: The Graph for reasoning

B)

Explain the concept of normalized cut in social network analysis. Calculate the normalized cut for the graph given below:

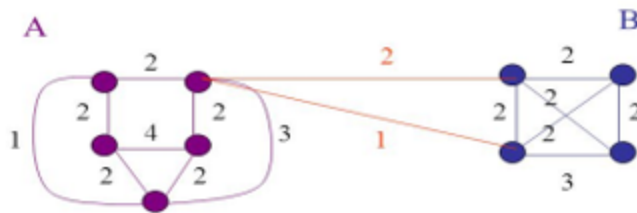


Figure 5: The Graph for reasoning

- 5) Describe the importance of influence-related statistics in social network analysis. How do these metrics assist in identifying key influencers? (5)

A)

- B) Describe the concept of an influence graph and a blog graph in social network analysis. How do these two types of graphs differ in their representation of relationships within a network? (5)

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